ExamGuard — Simple Step-by-Step User Manual

Beginner-Friendly Operational & Interface Guide with Real Application Screenshots

ExamGuard — Simple Step-by-Step User Manual

A Complete Beginner-Friendly Guide with Real Application Screenshots

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Welcome to ExamGuard

Welcome to **ExamGuard**! This user manual is designed for teachers, examination officers, and administrators who are using ExamGuard for the very first time.

You do **not** need any computer science or cybersecurity background to use this system. Every instruction in this guide is written in clear, everyday language with step-by-step actions and real screenshots from the live application.

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1. What is ExamGuard?

In many colleges and universities, examination question papers are emailed as attachments, shared on Google Drive, or carried on USB pen drives. These old methods are dangerous:

- Files can be forwarded or intercepted by unauthorized people.
- Papers can be leaked days before the exam takes place.
- Someone could change a number or formula inside the document, and nobody would know.
- USB drives can be lost or copied.

ExamGuard solves this completely:

1. **Instant Protection**: The moment a teacher uploads a question paper, the system locks it in a digital vault using military-grade encryption (AES-256).
2. **Digital Wax Seal**: The system stamps a unique digital seal on the file (an RSA-PSS digital signature). If anyone tampers with even a single dot, the seal breaks and warns you.
3. **Time-Locked Delivery**: People cannot download the paper whenever they want. The system keeps it locked until the official scheduled delivery date.
4. **Answer Key Protection**: Answer keys cannot be uploaded before an exam happens, preventing leaks.
5. **No Secrets in Plain Text**: The actual unencrypted PDF is never stored on disk. It is decrypted only in computer memory for the person authorized to view it.

2. How to Log In

2.1 Opening the Login Page

Open your web browser (Google Chrome, Microsoft Edge, Firefox, or Safari) and go to:

`https://examguardapp.vercel.app/login`

You will see the secure login screen:

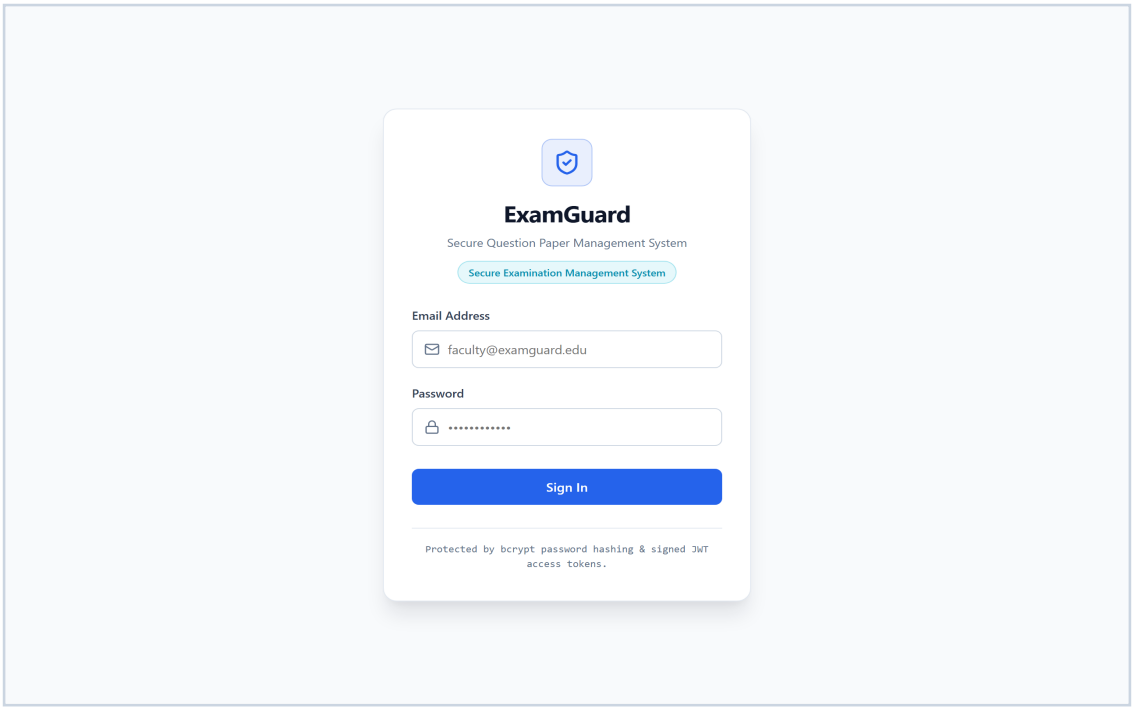


Figure 1: ExamGuard Secure Login Screen

Figure 1: ExamGuard Secure Login Screen.

- **What you are looking at**: The clean, dark-theme login portal of ExamGuard.
- **Key elements**:
- The **Email Address** box with the mail icon.
- The **Password** box with the lock icon.
- The blue **Sign In to Secure Workspace** button.
- The bottom demo card showing sample accounts.
- **What to do**: Click inside the Email field and type your institutional email address.
- **Expected result**: Your email appears in the box.

2.2 Entering Your Credentials

Type your registered institutional email address and your password.

The official demonstration accounts available on the system are:

- **Teacher**: `teacher@examguard.edu` | Password: `Teacher@ExamGuard2026`
- **Exam Coordinator**: `coordinator@examguard.edu` | Password: `Coord@ExamGuard2026`
- **Administrator**: `admin@examguard.edu` | Password: `Admin@ExamGuard2026`

Click the blue **Sign In to Secure Workspace** button.

2.3 What Happens When You Log In

When you click **Sign In**:

1. The server checks your email and verifies your password securely.
2. If your credentials are correct, you will see a green notice: *"Authentication successful! Redirecting..."*
3. The system automatically reads your assigned role and takes you directly to your workspace:

- Teachers go to `/teacher`.
- Coordinators go to `/exam-coordinator`.

- Administrators go to `/admin`.

4. Your login session stays active for **24 hours**. When you are done, click the **Sign Out** button in the top-right corner.

2.4 Handling Login Errors

If you type the wrong email or an incorrect password, ExamGuard displays a clear red alert message:

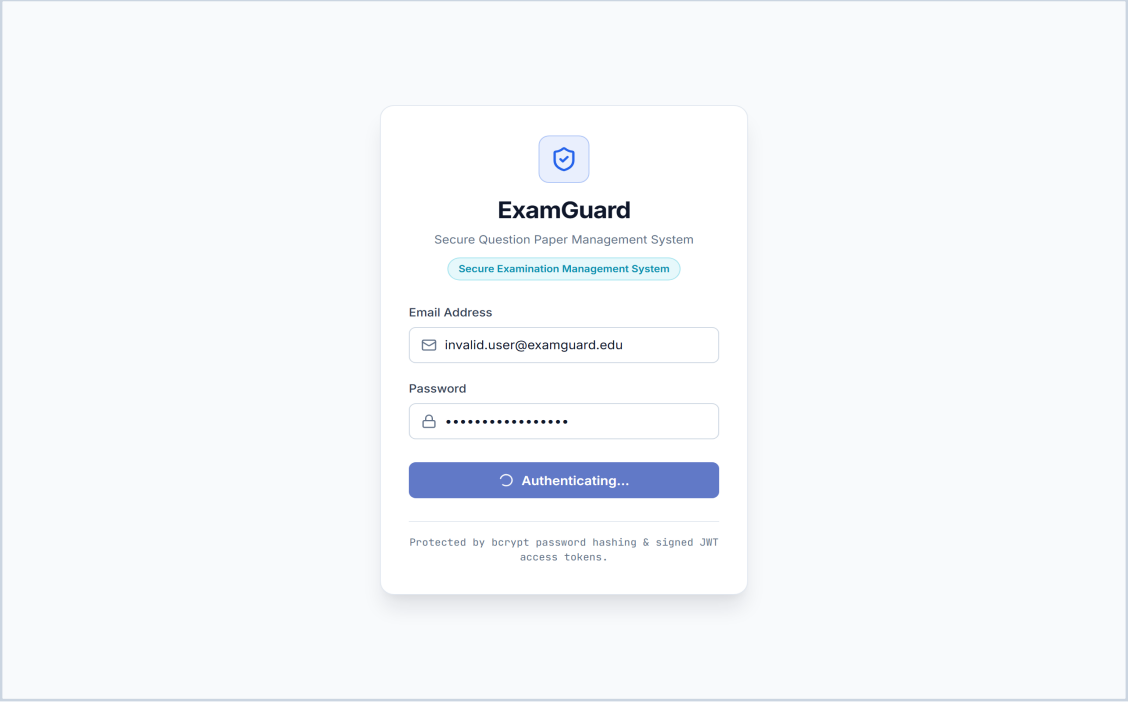


Figure 2: Login Validation Error Screen

Figure 2: Login Validation Error Screen.

- **What you are looking at**: The red warning banner that appears when login fails.
- **Key elements**: The red alert box with the text *"Invalid email or password"*.
- **What to do**: Double-check your spelling and Caps Lock key. Re-type your email and password, then click Sign In again.
- **Expected result**: Once corrected, the red banner disappears and you are logged in.

3. Understanding the Three Roles

ExamGuard uses **Role-Based Access Control**. This means what you can see and do depends on your job:

```
+-------------------------------------------------------------------------------+
| USER ROLES IN EXAMGUARD                                                       |
+-----------------------+-------------------------+-----------------------------+
| TEACHER               | EXAM COORDINATOR        | ADMINISTRATOR               |
|-----------------------+-------------------------+-----------------------------|
| - Uploads papers      | - Receives papers       | - Creates user accounts     |
| - Generates papers    | - Verifies signatures   | - Activates / deactivates users |
| - Reviews assignments | - Downloads on exam day | - Resets forgotten passwords |
| - Uploads answer keys | - Views answer keys     | - Reviews security audit logs |
| - Deletes own drafts  | - Views exam rosters    | - CANNOT see or decrypt papers |
+-------------------------------------------------------------------------------+
```

4. Teacher Guide — Complete Walkthrough

4.1 What Teachers Do

As a faculty member or teacher, you author examination question papers. In ExamGuard, you can:

- Upload finalized PDF question papers you prepared offline.
- Use the built-in wizard to assemble a fresh question paper from an institutional question bank.
- Check the cryptographic integrity of your papers anytime.
- Assign papers to reviewers or exam coordinators.
- Submit answer keys during the post-exam submission window.
- Safely delete drafts you no longer need.

4.2 The Teacher Dashboard

Log in with:

- **Email**: `teacher@examguard.edu`
- **Password**: `Teacher@ExamGuard2026`

You will arrive at your home dashboard:

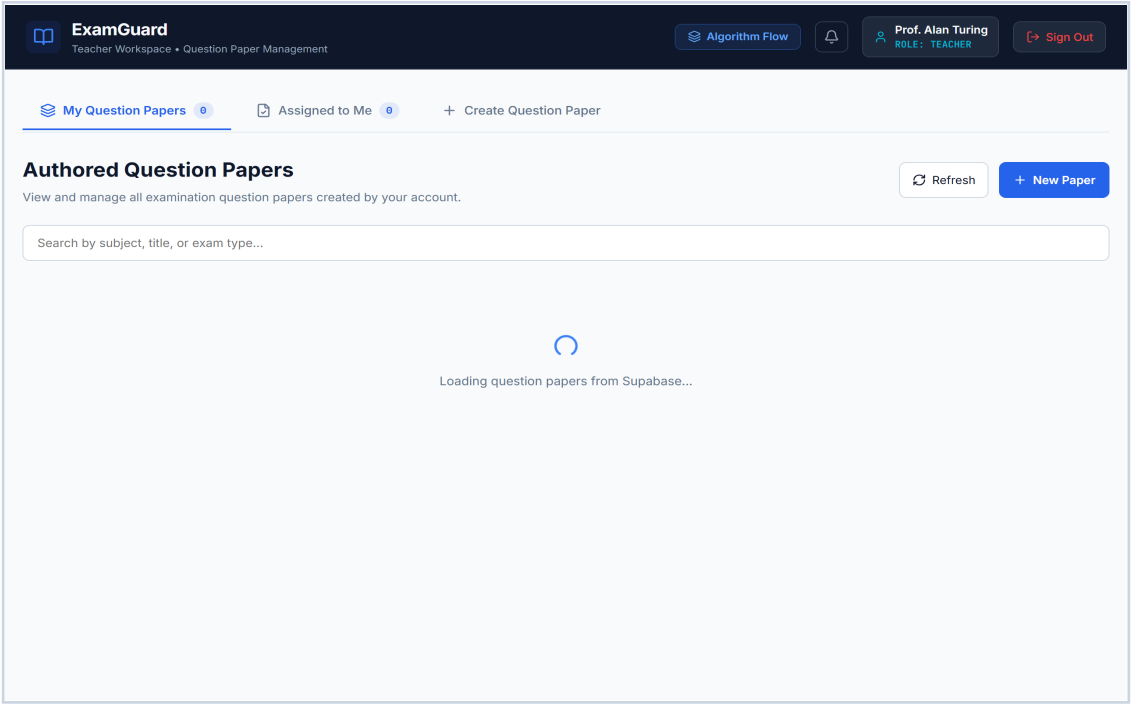


Figure 3: Teacher Dashboard Overview

Figure 3: Teacher Dashboard Overview.

- **What you are looking at**: The Teacher Dashboard showing authored question papers.
- **Key elements**:
- **Top Navigation Bar**: Displays the ExamGuard logo, your name (*Prof. Alan Turing*), your role badge (`TEACHER`), the notification bell, and the **Sign Out** button.
- **Action Ribbon**: The **View Algorithm Flow** button (top right) and the three main workspace tabs:

1. **My Question Papers**: Papers you authored.
2. **Assigned to Me**: Papers authored by other teachers sent to you for review.
3. **Create Question Paper**: Where you upload or generate new papers.

- **Question Papers Table**: Shows Title, Subject, Semester, Exam Date, Scheduled Delivery Date, and Status badge.
- **Action Buttons in Table**: Details (eye icon), Verify Signature (shield), Download, Answer Key, and Delete (trash).

4.3 Viewing Your Own Authored Papers

Click the **My Question Papers** tab.

- You will see a list of every paper you created.
- The status pill indicates the paper's progress:

- `SCHEDULED`: The paper is safely encrypted and waiting for its delivery date.
- `DELIVERED`: The delivery date has arrived; the paper is ready.
- `SUBMITTED`: Upload completed.
- As the author, you have **instant download rights** to your own papers. You do not have to wait for the delivery date to review your work.

4.4 Viewing Papers Assigned to You

Click the **Assigned to Me** tab:

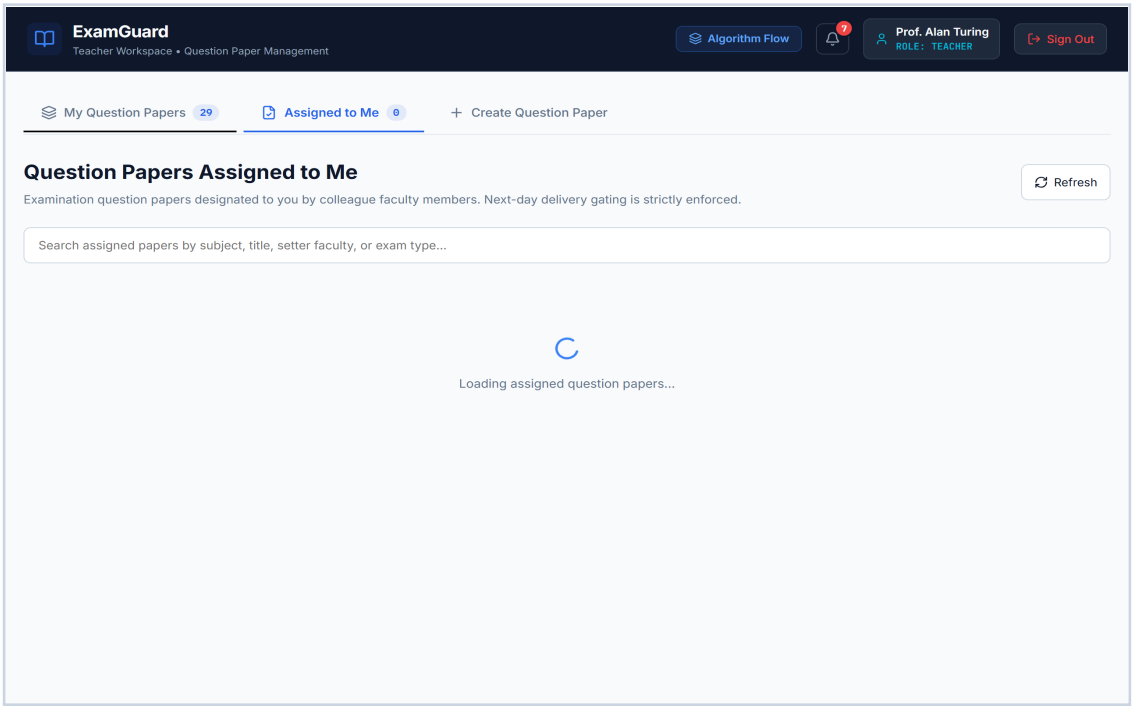


Figure 4: Assigned to Me Papers Tab

Figure 4: Assigned to Me Papers Tab.

- **What you are looking at**: Papers authored by colleagues that were assigned to you for co-review.
- **Key elements**:
- The table columns showing the Author name, Scheduled Delivery Date, and Delivery Status.
- The disabled **Download** button with a lock icon.
- **Important Rule**: For assigned papers, you cannot download the PDF until the scheduled delivery date arrives. This prevents early leakage. You can, however, verify the digital signature immediately.

4.5 How to Upload an Existing PDF Question Paper (Option A)

If you already typed and saved your question paper as a PDF on your computer, follow these steps:

1. Click the **Create Question Paper** tab.
2. Click the **Upload Existing PDF** card.
3. The upload form opens:

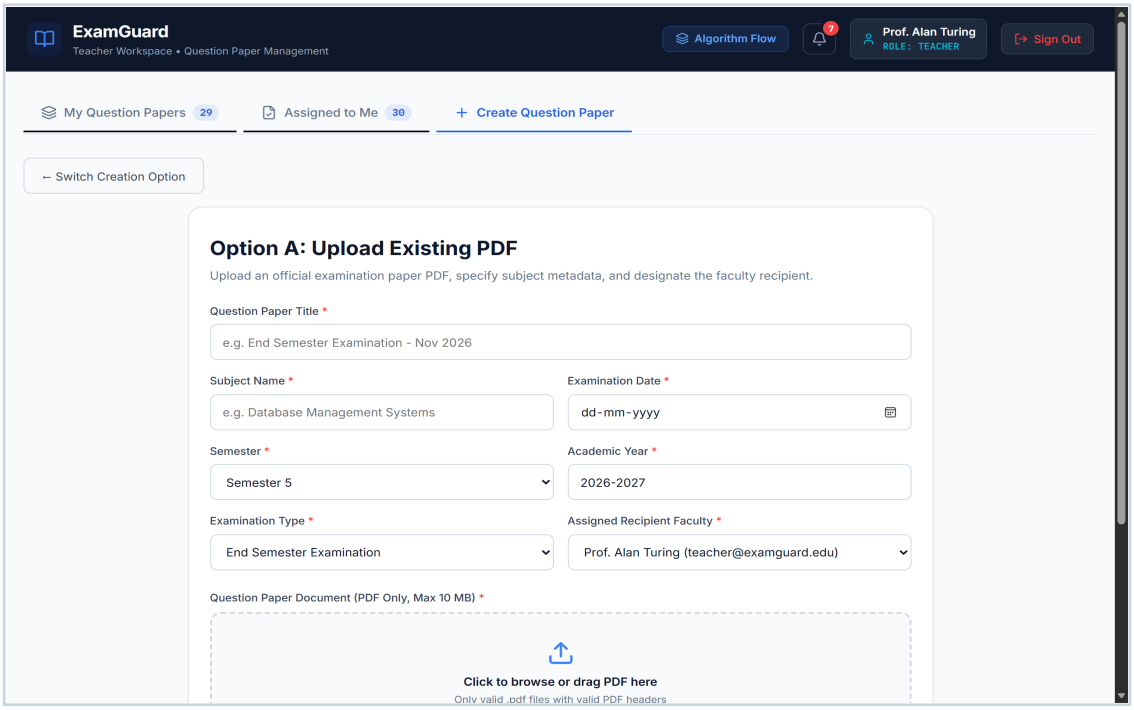


Figure 5: Teacher Question Paper Upload Form

Figure 5: Teacher Question Paper Upload Form (Option A).

- **What you are looking at**: The manual PDF submission form.
- **Step-by-Step Form Inputs**:

1. **Question Paper Title**: Enter the formal name (e.g., `Data Structures & Algorithms Final Exam`).
2. **Subject**: Enter the course code or name (e.g., `CS-301 Computer Networks`).
3. **Semester**: Select your semester from the dropdown (e.g., `Semester 5`).
4. **Academic Year**: Type the session (e.g., `2026-2027`).
5. **Exam Type**: Choose `End Semester Examination`, `In-Semester Test`, or `Class Test`.
6. **Exam Date**: Click the calendar picker and select the future date of the exam.
7. **Assigned Recipient**: Pick the reviewer or exam coordinator from the dropdown.
8. **File Upload**: Drag and drop your `.pdf` file into the blue dashed box, or click to browse.
4. Click the blue **Submit Question Paper** button.
5. **What happens next**:

- The server validates your PDF file.
- It generates a unique 256-bit AES encryption key.
- It encrypts the PDF so nobody can read it in storage.
- It seals it with a digital signature.
- It sets the delivery date to tomorrow (`today + 1 day`).
- A green confirmation alert appears, and your paper is listed under **My Question Papers**.

4.6 How to Generate a Question Paper via the Wizard (Option B)

ExamGuard can also automatically build a complete, balanced examination paper set for you based on a university syllabus blueprint.

1. Click **Create Question Paper** $\rightarrow$ Click **Generate via Wizard**.

2. **Step 1: Exam Details & Blueprint**:

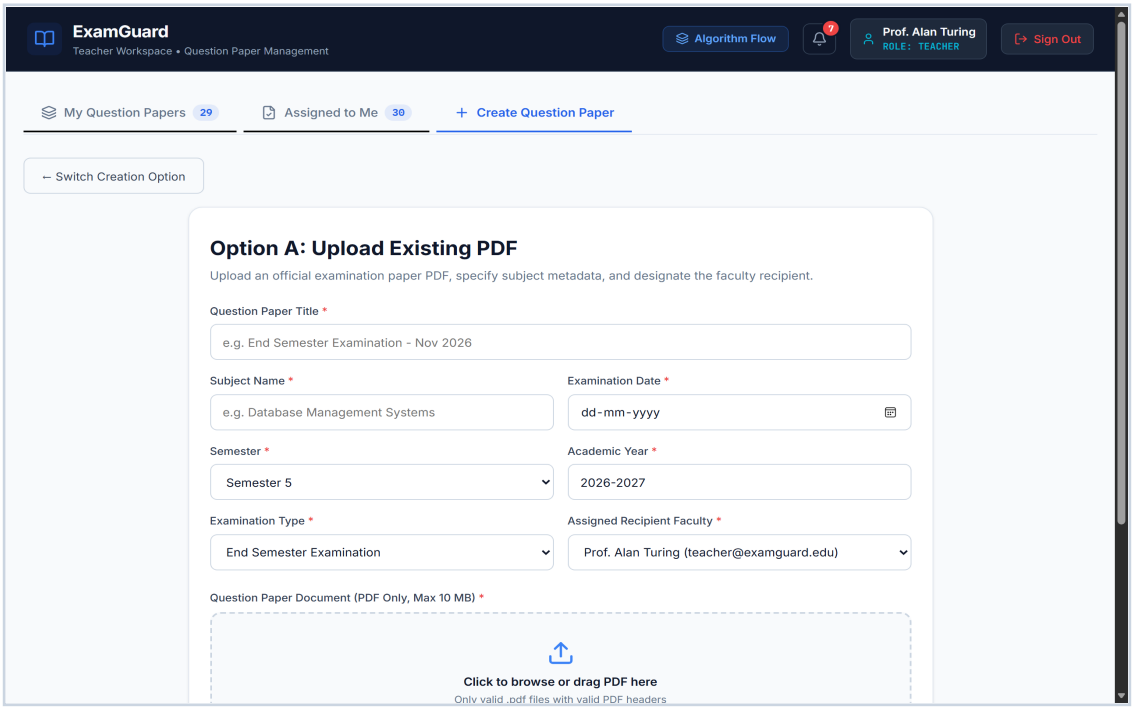


Figure 6: Question Paper Generator Wizard — Step 1: Blueprint

Figure 6: Question Paper Generator Wizard — Step 1: Blueprint.

- **What you are looking at**: Step 1 of the 5-step wizard.
- **What to do**: Review the Course, Semester, Pattern, Subject Code (`310243`), Total Marks (`70`), and Duration (`2.5 Hours`). Check the instructions.
- Click **Next: Question Bank & Units**.

3. **Step 2: Question Bank & Units**:

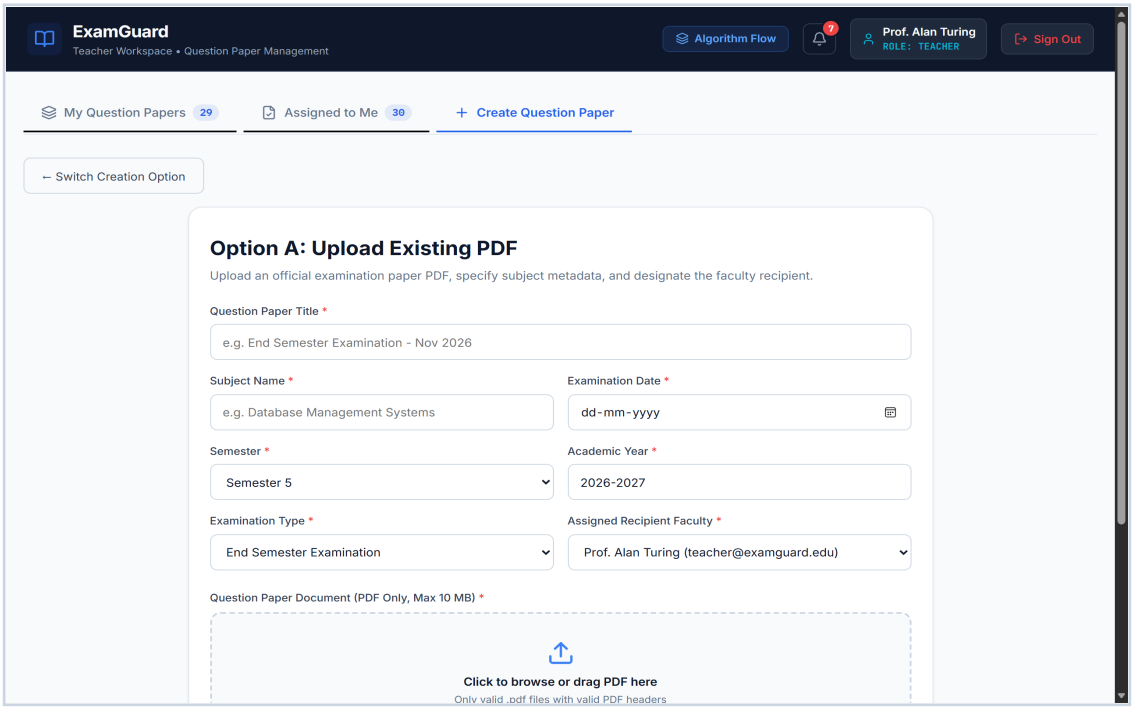


Figure 7: Question Paper Generator Wizard — Step 2: Question Bank

Figure 7: Question Paper Generator Wizard — Step 2: Question Bank.

- **What you are looking at**: The list of approved questions in the question bank, organized by unit number, difficulty, and marks.
- **What to do**: Review available questions. You can click **Add Question** to input your own custom question or click **Seed Default SPOS Question Bank** if the bank is empty.
- Click **Next: Feasibility & Set Generation**.

4. **Step 3: Feasibility Analysis**:

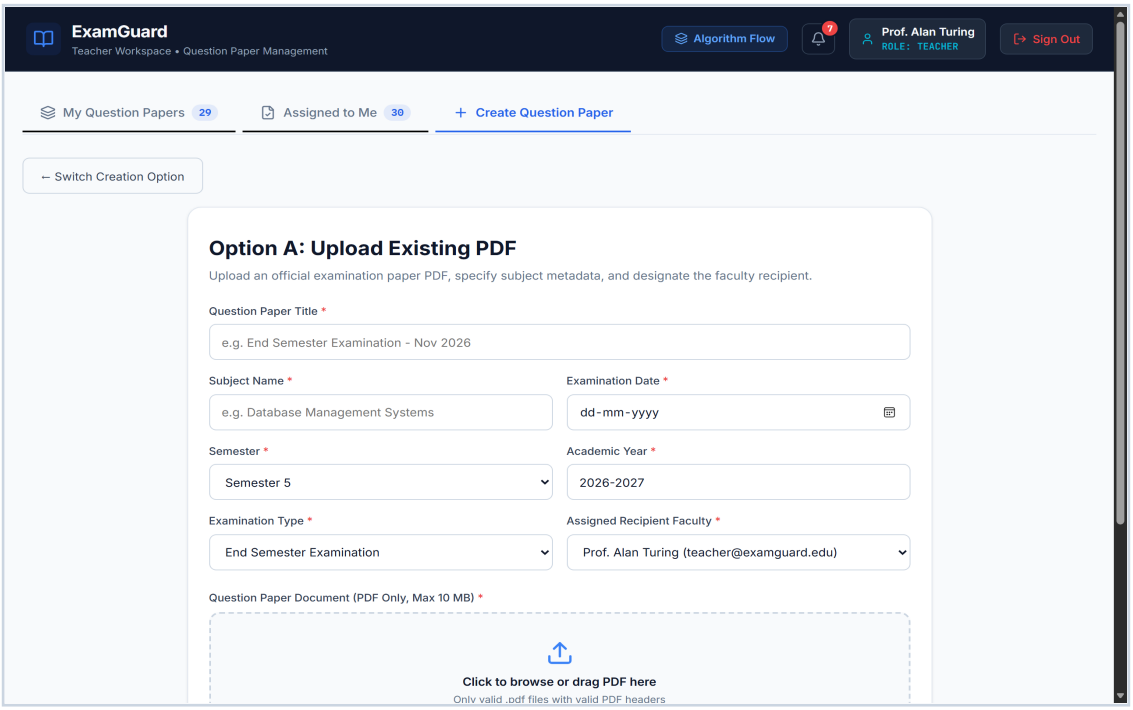


Figure 8: Question Paper Generator Wizard — Step 3: Feasibility

Figure 8: Question Paper Generator Wizard — Step 3: Feasibility.

- **What you are looking at**: The feasibility check.
- **What to do**: The system calculates whether there are enough questions to create **Set A** and **Set B** without exceeding the 40% overlap limit.
- Click **Generate Sets Now**.

5. **Step 4: Preview & In-Place Swap**:

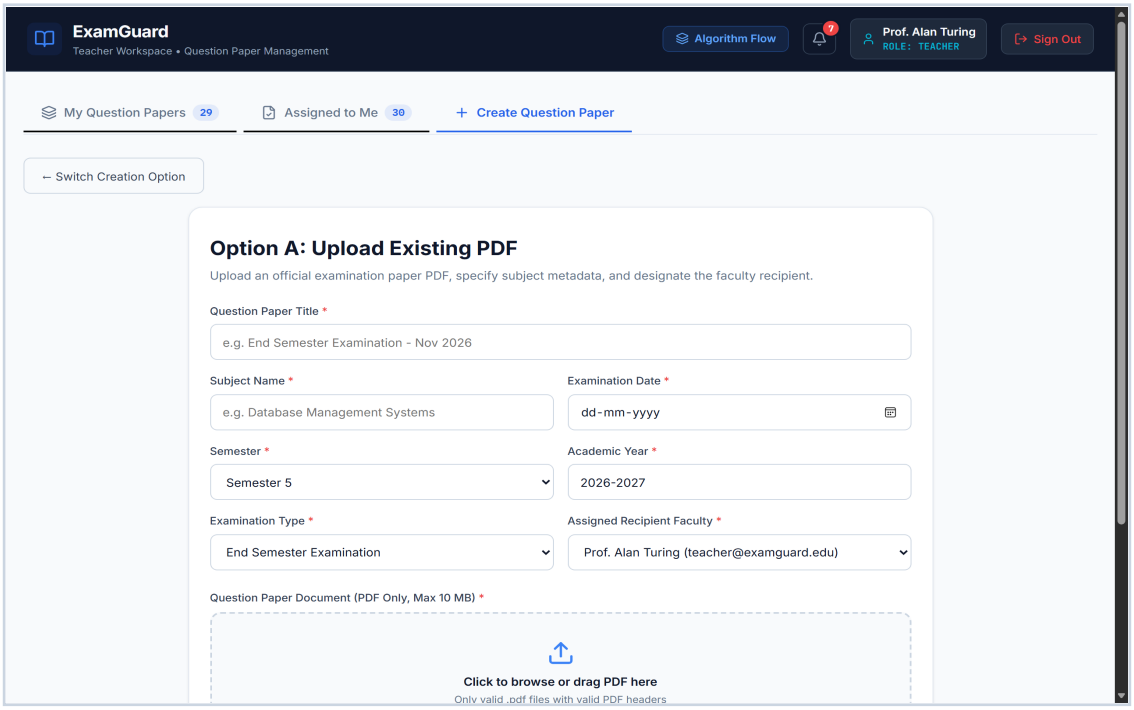


Figure 9: Question Paper Generator Wizard — Step 4: Preview & Swap.

Figure 9: Question Paper Generator Wizard — Step 4: Preview & Swap.

- **What you are looking at**: The generated university-style paper layout.
- **What to do**:
- Toggle between **Set A** and **Set B**.
- Notice the formal university header, Seat Number box, and `[OR]` question groups.
- If you dislike a question, click **Swap Question**. A list of alternative questions with the exact same marks and difficulty opens. Pick an alternative, and the marks total updates automatically!
- Click **Next: Final Review & Submit**.

6. **Step 5: Approve & Submit**:

- Select your recipient and exam date.
- Check the confirmation box: *"I hereby review, approve, and authorize the generated paper."*
- Click **Approve & Submit to Secure Pipeline**. The system renders the paper into a publication-grade PDF and encrypts it immediately.

4.7 How to Verify That a Paper Has Not Been Modified

You can verify the digital signature of any paper at any time without decrypting it:

1. Click the **View Algorithm Flow** button on the dashboard to see how the security works:

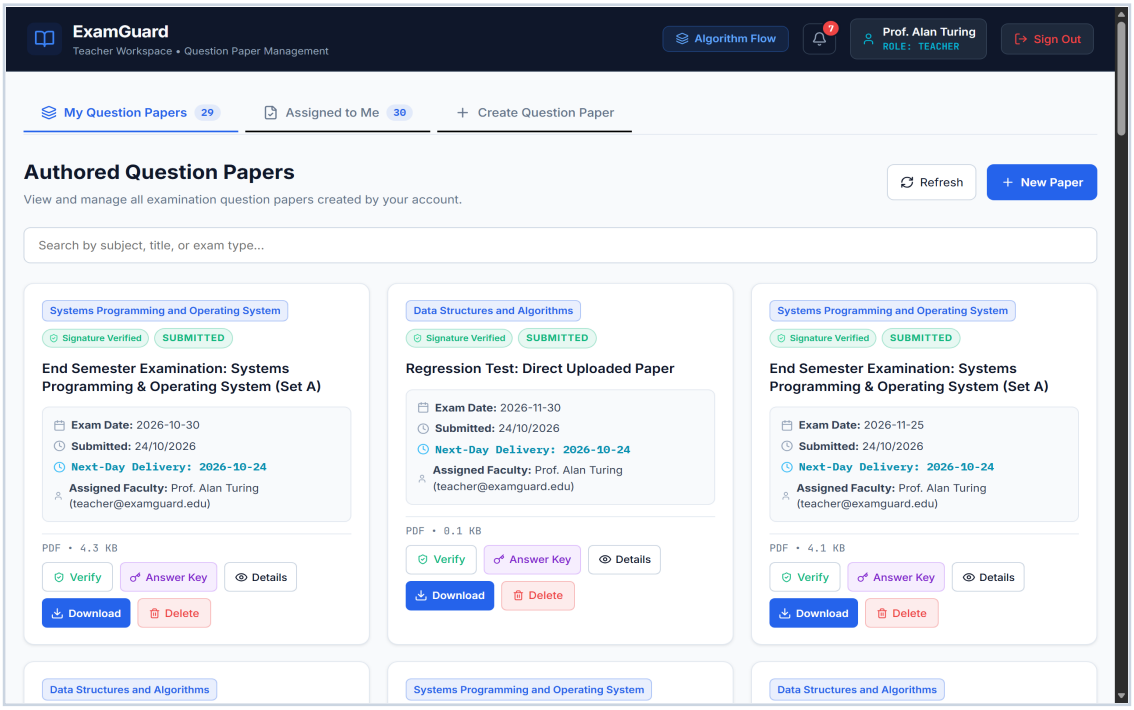


Figure 10: Cryptographic Algorithm Flow Modal.

Figure 10: Cryptographic Algorithm Flow Modal.

- **What you are looking at**: The 11-step visual diagram explaining the cryptographic pipeline.

2. In the paper table, click the blue **Verify Signature** (shield) button next to any paper.
3. **What happens**: The system checks the digital signature against the stored file. If the file is untouched, a green confirmation badge appears: `Verified (RSA-PSS-SHA256)`. If anyone had tampered with the file, a prominent red warning would appear.

4.8 How to Download a Question Paper (and Scheduled Delivery Dates)

- In the table, click the green **Download** button.
- **For Authoring Teachers**: The paper downloads immediately as a decrypted PDF.
- **For Reviewers & Coordinators**:
- If today's date is **before** the scheduled delivery date, the button is disabled and shows a padlock. You will see a tooltip: *"Scheduled for delivery on [Date] and cannot be downloaded yet."*
- On or after the delivery date, the button turns green and you can download the decrypted PDF.

4.9 How to Upload and Manage Post-Exam Answer Keys

To make sure answer keys are never leaked before students take the test, ExamGuard enforces a **strict submission window**:

1. In your **My Question Papers** list, click the **Answer Key** button next to a paper.
2. The Answer Key Management Modal opens:

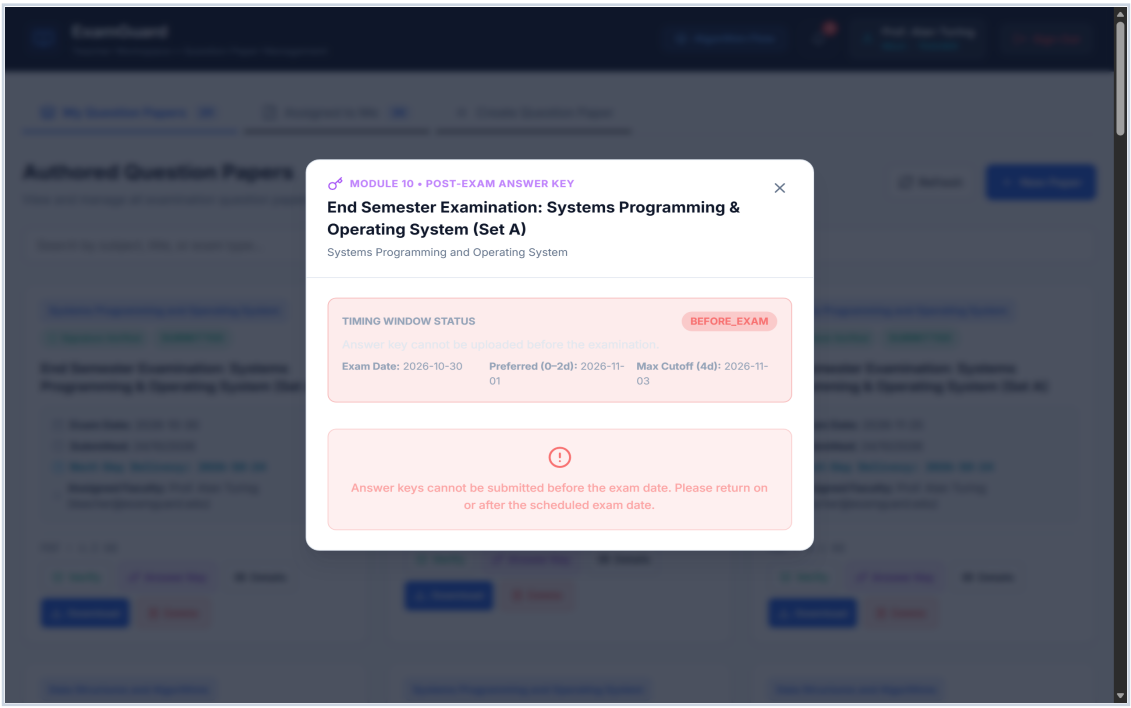


Figure 11: Post-Exam Answer Key Modal

Figure 11: Post-Exam Answer Key Modal.

- **What you are looking at**: The Answer Key status and timing window.
- **Timing Window Rules**:
- **Before Exam Date**: Status is `BEFORE_EXAM` in yellow. Upload is locked.
- **Days 0–2 After Exam**: Status is `WINDOW_OPEN_PREFERRED` in green. Upload is open!
- **Days 3–4 After Exam**: Status is `WINDOW_OPEN_EXTENDED` in amber. Upload is open!
- **Day 5+ After Exam**: Status is `WINDOW_EXPIRED` in red. Upload is closed.
- **Uploading**: When the window is open, choose your solution PDF and click **Upload Answer Key**.
- The answer key is encrypted and signed with its own cryptographic envelope.

4.10 How to Delete an Authored Paper Safely

If you uploaded a draft paper by mistake and want to remove it:

1. In the paper table, click the red **Delete (Trash)** icon.
2. The confirmation modal opens:

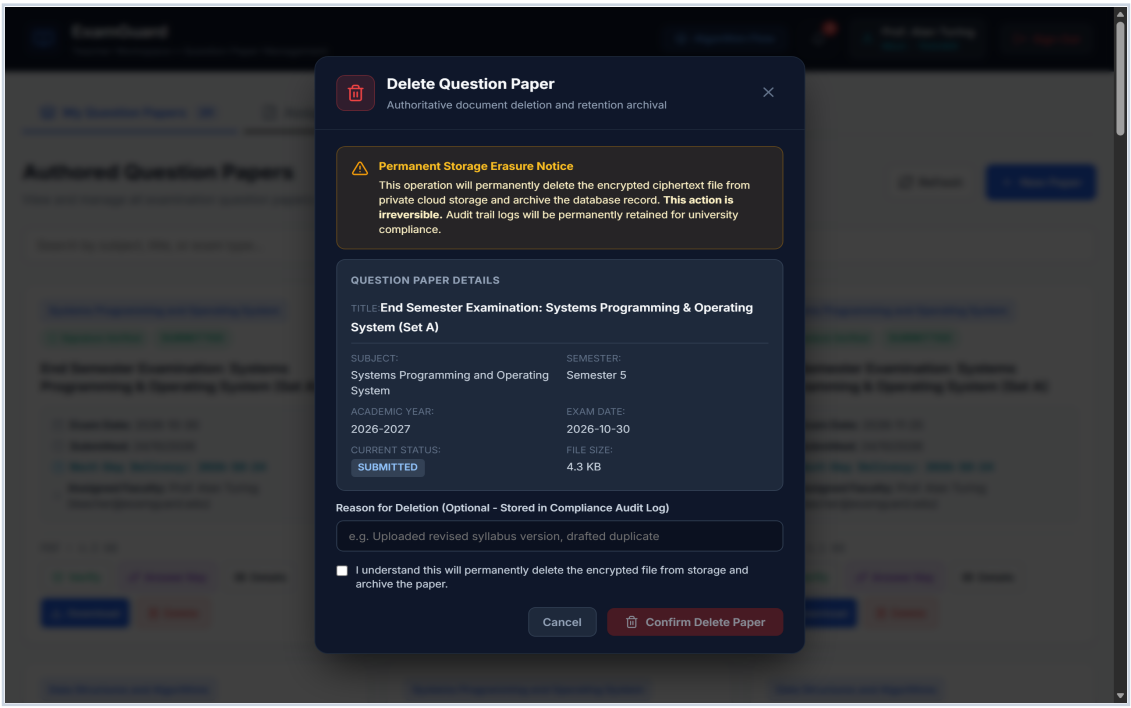


Figure 12: Delete Question Paper Modal

Figure 12: Delete Question Paper Modal.

- **What you are looking at**: The deletion safety confirmation box.
- **What to do**: Read the warning. You may optionally enter a reason in the text box.
- Click **Permanently Delete**.
- **Safety Lock**: If the paper has already reached its delivery date or was delivered to a coordinator, the system **blocks** deletion to protect institutional records. If it is still a pending draft, the file is erased from cloud storage and soft-deleted.

4.11 What Teachers Cannot Do

- Teachers cannot download assigned papers before the scheduled delivery date.
- Teachers cannot upload answer keys before the exam date.
- Teachers cannot delete papers that have already been delivered.
- Teachers cannot view or edit papers authored by other faculty unless explicitly assigned.
- Teachers cannot access administrator user management or system audit logs.

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5. Exam Coordinator Guide — Complete Walkthrough

5.1 What Exam Coordinators Do

Exam Coordinators oversee paper distribution on examination day. In ExamGuard, coordinators:

- View all question papers assigned to their examination center.
- Verify cryptographic signatures in advance to ensure no files were corrupted or altered.
- Download decrypted examination papers once the scheduled delivery date arrives.
- Access post-exam answer keys for evaluation.

5.2 The Coordinator Dashboard

Log in with:

- **Email**: `coordinator@examguard.edu`
- **Password**: `Coord@ExamGuard2026`

You will see the Coordinator workspace:

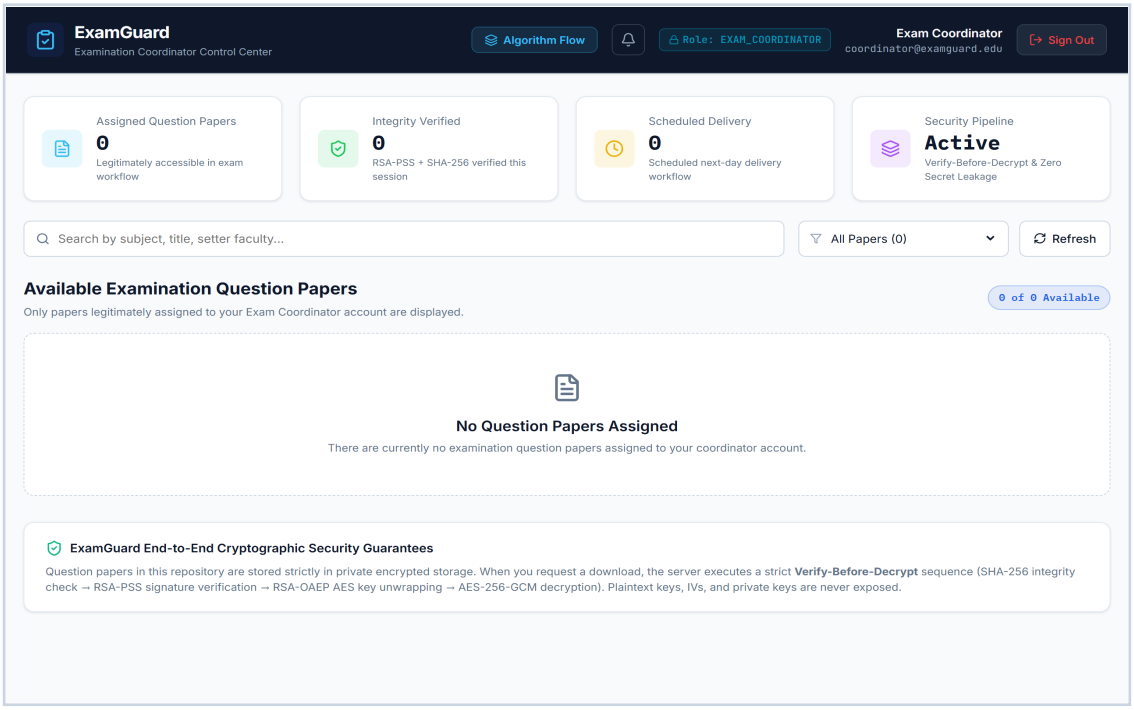


Figure 13: Exam Coordinator Dashboard Overview

Figure 13: Exam Coordinator Dashboard Overview.

- **What you are looking at**: The Exam Coordinator repository.
- **Key elements**:
- **Summary Metric Cards**: Shows Total Assigned Papers, Verified Signatures count, and Ready for Delivery count.
- **Search & Filter Bar**: Type any subject name or choose a status (`ALL`, `SCHEDULED`, `DELIVERED`).
- **Assigned Papers Table**: Shows Paper Title, Subject, Authoring Teacher, Delivery Date, and Status.
- **Action Buttons**: Details (eye icon), Verify Signature (shield), Download, and View Answer Key.

5.3 Finding Papers with Search and Filters

- In the search bar at the top of the table, type any course name (e.g., `SPOS` or `Networks`). The table filters instantly.
- Click the Status dropdown to view only papers that are `SCHEDULED` or `DELIVERED`.

5.4 Viewing Paper Details

Click the **Details (Eye)** icon next to any paper:

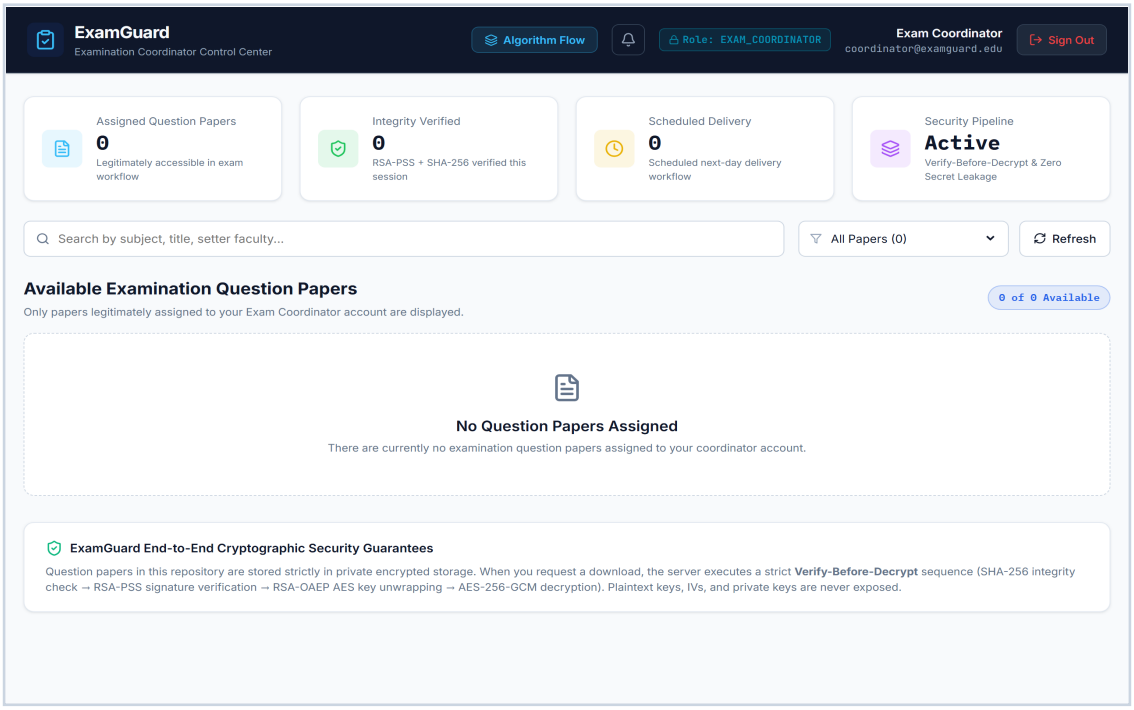


Figure 14: Coordinator Paper Details Modal

Figure 14: Coordinator Paper Details Modal.

- **What you are looking at**: The complete metadata inspection sheet.
- **Information shown**: System UUID, Paper Title, Subject, Semester, Author Name, Exam Date, Scheduled Delivery Date, and File Size.
- Click **Close** to return to the table.

5.5 Checking Paper Signatures

1. In the table, click the blue **Verify Signature** button.
2. The server checks the cryptographic digest against the digital signature stored in the database.
3. A green confirmation badge confirms that the file is authentic and has not been tampered with.

5.6 Downloading the Paper on Exam Day

- When the scheduled delivery date arrives (e.g., the morning of the exam), the **Download** button turns bright green.
- Click **Download**.
- The server performs the 5-step verification pipeline:

1. Verifies the SHA-256 hash.
2. Verifies the RSA-PSS signature.
3. Unwraps the AES key.
4. Decrypts the paper.
5. Clears the key from server memory.

- The authenticated PDF opens in your browser, ready for authorized printing in the secure exam cell.

5.7 Accessing Post-Exam Answer Keys

- Once an authoring teacher uploads the answer key after the exam, an **Answer Key** button appears in the coordinator table.
- Click **View Answer Key** to inspect verification details or download the solution guide.

5.8 What Coordinators Cannot Do

- Coordinators cannot download papers before the scheduled delivery date.
- Coordinators cannot view or download papers that were not explicitly assigned to them.
- Coordinators cannot upload or delete question papers.
- Coordinators cannot access the admin user management panel.

6. Administrator Guide — Complete Walkthrough

6.1 What Administrators Do

System Administrators manage institutional accounts and monitor security compliance. Administrators:

- Provision user accounts for teachers and coordinators.
- Activate or deactivate accounts when faculty join or leave.
- Assign system roles (`TEACHER`, `EXAM_COORDINATOR`, `ADMIN`).
- Assist with password resets when users forget their credentials.
- Inspect the immutable security audit trail.
- **Important Security Principle**: Administrators have **no access** to examination papers or encryption keys. They manage people and security, not confidential exam content.

6.2 The Administrator Dashboard

Log in with:

- **Email**: `admin@examguard.edu`
- **Password**: `Admin@ExamGuard2026`

You will see the Administrator portal:

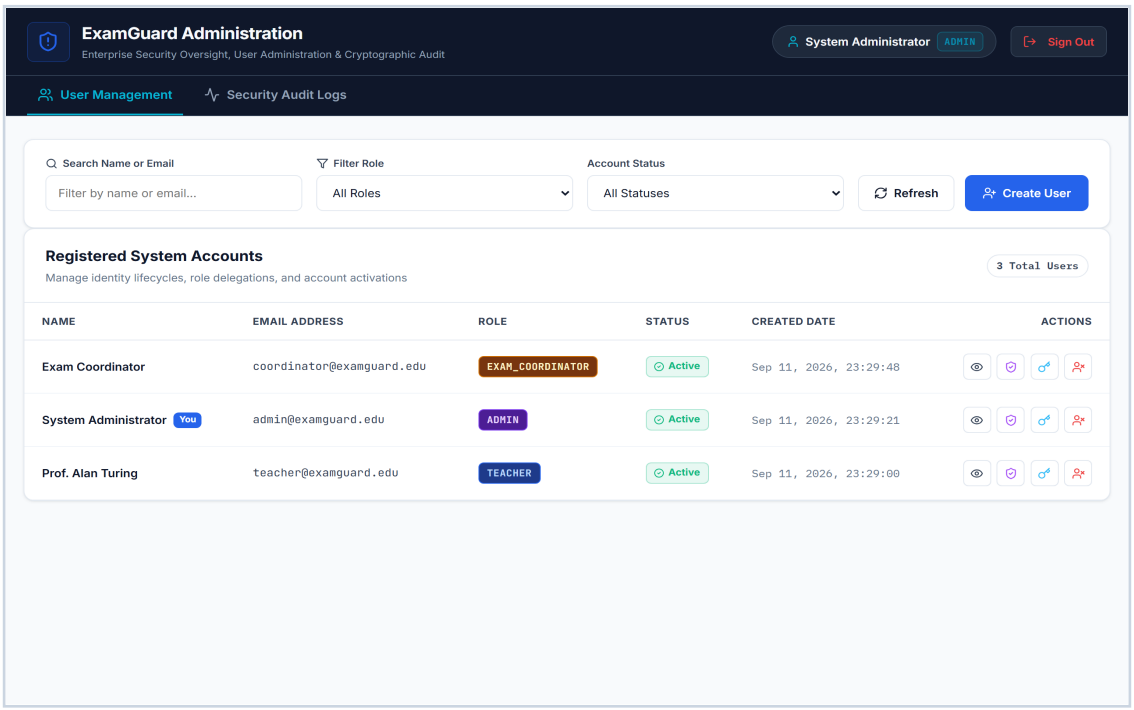


Figure 15: Administrator User Management Dashboard

Figure 15: Administrator User Management Dashboard.

- **What you are looking at**: The User Management tab in `/admin`.
- **Key elements**:
- **Top Tabs**: Switch between **User Management** and **Audit Logs**.
- **Search & Filters**: Search users by name or email; filter by Role or Status (`ACTIVE` / `INACTIVE`).

- **Create New User Button**: Blue button in the upper right.
- **User Directory Table**: Displays Name, Email, Role badge, Status pill, and Action controls.
- **Action Controls in Table**:
- **Role Selector**: Change a user's role.
- **Deactivate / Activate Button**: Toggle access on or off.
- **Reset Password (Key Icon)**: Reset a user's password.

6.3 Creating a New User Account

1. Click the blue **Create New User** button.
2. The user creation form opens:

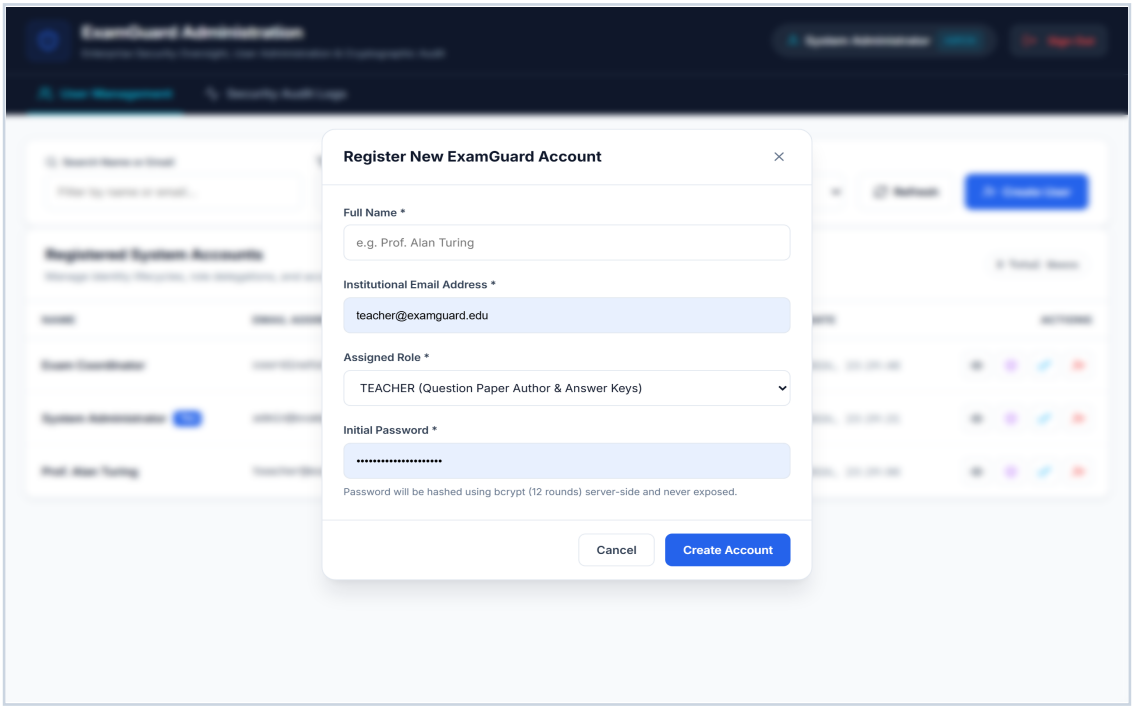


Figure 16: Admin Create User Modal

Figure 16: Admin Create User Modal.

- **What you are looking at**: The account provisioning dialogue.
- **Form Fields**:
- **Full Name**: Enter the user's name (e.g., `Dr. Jane Doe`).
- **Email Address**: Enter their institutional email (e.g., `jane.doe@examguard.edu`).
- **Initial Password**: Enter a temporary password (must be at least 8 characters with at least 1 letter and 1 number).
- **Role**: Select `TEACHER` or `EXAM_COORDINATOR`.
3. Click **Create User**.
4. The server hashes the password with bcrypt (12 rounds) and creates the account.

6.4 Activating or Deactivating an Account

- To disable a user (for example, if an instructor is on sabbatical or leaves the institution), click **Deactivate** next to their name.
- The status pill changes to red `INACTIVE`.
- **Immediate Effect**: The user is immediately blocked from logging in. If they have an active session, their very next click will return an access denied error.
- To re-enable the user, click **Activate**.

6.5 Changing a User's Role

- In the user table, find the **Role** column dropdown.
- Select the new role (e.g., change from `TEACHER` to `EXAM_COORDINATOR`).
- Confirm the change in the prompt. The user's permissions update instantly.

6.6 Resetting a User's Password

If a faculty member forgets their password:

1. Click the **Key Icon (Reset Password)** next to their name.
2. Type a temporary password matching complexity rules (at least 8 characters, letters and numbers).
3. Click **Save Password**. The system hashes it and updates the account.

6.7 Viewing the Security Audit Logs

Click the **Audit Logs** tab at the top of the dashboard:

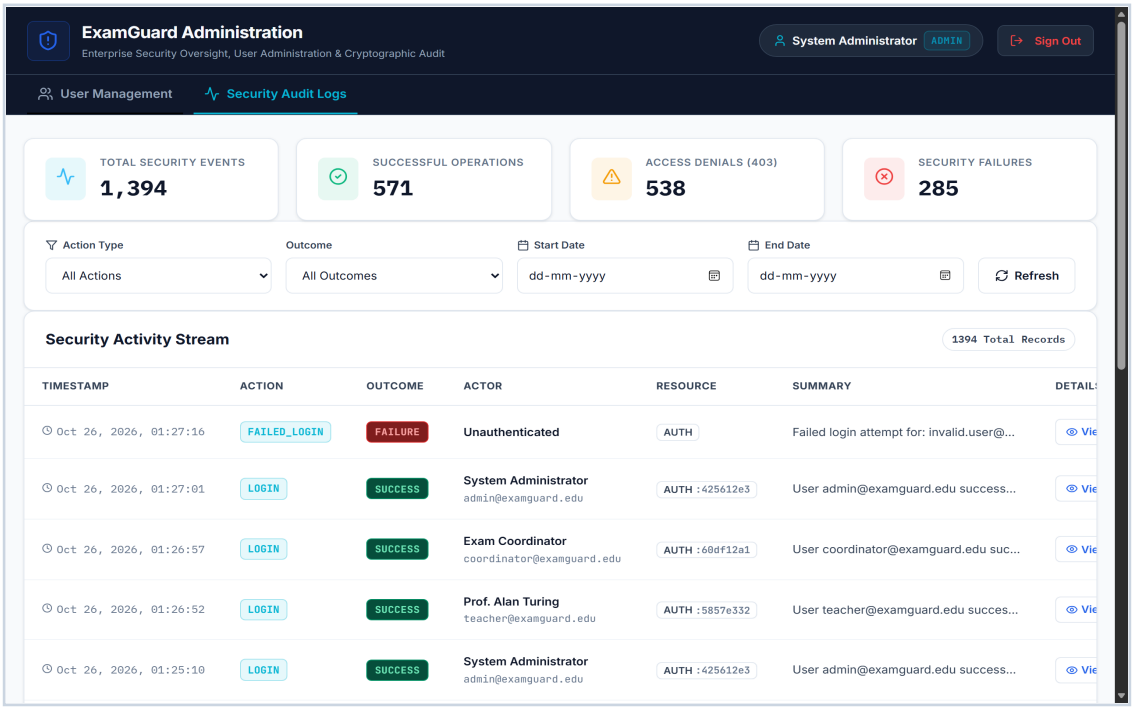


Figure 17: Administrator Audit Logs Dashboard

Figure 17: Administrator Audit Logs Dashboard.

- **What you are looking at**: The tamper-evident security audit trail.
- **Key elements**:
- **Four Metric KPI Cards**:
- **Total Events**: Total number of recorded actions.
- **Success Count**: Successful logins, uploads, and downloads.
- **Failure Count**: System or processing errors.
- **Denied Count**: Blocked attempts, IDOR attempts, or expired window attempts.
- **Filter Toolbar**: Filter events by Action type (e.g., `LOGIN`, `QUESTION_PAPER_SUBMITTED`, `DOWNLOAD`), Outcome (`SUCCESS`, `DENIED`), or Date Range.
- **Audit Log Table**: Lists Event Action, Outcome pill (green/red), Actor email, Target Resource, Timestamp, and a **View Details** button.

6.8 Inspecting Audit Event Details

Click the **View Details (Eye)** icon next to any log entry:

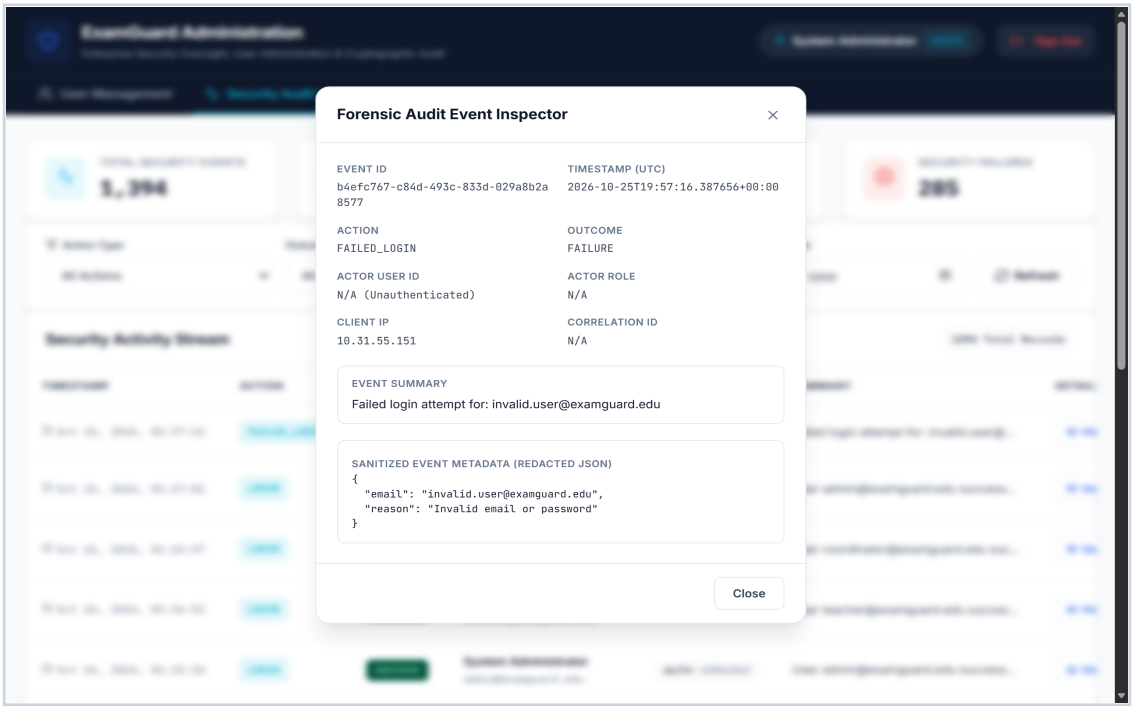


Figure 18: Audit Log Sanitized JSON Inspection Modal

Figure 18: Audit Log Sanitized JSON Inspection Modal.

- **What you are looking at**: The full event inspection modal.
- **Key elements**:
- The Event Action, Outcome, User ID, IP address, and Timestamp.
- The formatted JSON payload box.
- **Automatic Redaction**: Notice that sensitive information is automatically masked:
- `"password": "[REDACTED]"`
- `"token": "[REDACTED]"`
- `"aesKey": "[REDACTED]"`
- `"storagePath": "[REDACTED]"`
- Secrets are never leaked into the audit logs.
- Click **Close** when finished.

6.9 Important Administrator Safety Rules

ExamGuard has built-in safety guardrails to protect the system:

1. **No Self-Lockout**: You cannot deactivate your own administrator account.
2. **No Self-Demotion**: You cannot change your own role away from `ADMIN`.
3. **Last Admin Protection**: You cannot deactivate or demote the last remaining active administrator account. This ensures the system can never become unmanaged.
4. **Immutable Logs**: Audit logs cannot be edited or deleted. There are no delete buttons on the audit trail.

---

7. Simple Flowcharts of Daily Workflows

7.1 Login & Dashboard Access

```
[ User enters Email & Password ]
               %
               %¼
   [ Server checks credentials ]
               %
      % % % % % % % % %4% % % % % % % % %
      %  Valid           %  Invalid
      %¼                 %¼
[ Issue 24-hr Token ]  [ Show Red Error Banner ]
      %
      % % % % % % % %
```

```
           [ Question Paper Submitted on Day 0 ]
                              %
                              %¼
           [ Server locks Delivery Date to Day 1 ]
                              %
             % % % % % % % % % % % % % % %4% % % % % % % % % % % % % % %
             %                                  %
             %¼                                 %¼
    [ Before Delivery Date ]       [ On or After Delivery Date ]
             %                                  %
             %¼                                 %¼
- Authoring Teacher CAN download   - Authoring Teacher CAN download
- Coordinator download is LOCKED   - Coordinator download is UNLOCKED
- Lock icon shown in table         - Green Download button active
- Inline signature verification OK - Verify-Before-Decrypt streams PDF
```

7.5 Coordinator Download Workflow

```
[ Coordinator clicks "Download" on Exam Day ]
                       %
                       %¼
[ 1. Check SHA-256 fingerprint: Has file changed? ] % %  Fail % % > [ STOP (Alert) ]
                       %  Pass
                       %¼
[ 2. Check Digital Signature: Is seal authentic? ] % % %  Fail % % > [ STOP (Alert) ]
                       %  Pass
                       %¼
[ 3. Unwrap AES key using server private key ] % % % % % % %  Fail % % > [ STOP (Alert) ]
                       %  Pass
                       %¼
[ 4. Decrypt PDF bytes and verify Auth Tag ] % % % % % % % % %  Fail % % > [ STOP (Alert) ]
                       %  Pass
                       %¼
[ 5. Wipe temporary key from memory buffer ]
                       %
                       %¼
[ Stream clean, authenticated PDF to user's screen! ]
```

7.6 Answer Key Submission Window

```
      Exam Day               Exam Day + 2 Days          Exam Day + 4 Days
% % % % % % % % % %<% % % % % % % % % % % % % % %
```

Q: Can an Exam Coordinator download a paper 3 days before the exam to print it?

A: No. ExamGuard enforces scheduled delivery date gating. Papers can only be downloaded on or after their server-calculated scheduled delivery date.

Q: What if I uploaded the wrong PDF by mistake?

A: If the paper has not yet been delivered to an exam coordinator, you can click the red **Delete (Trash)** button to permanently erase the encrypted file and upload the correct one.

Q: What happens if someone hacks into the cloud storage bucket directly?

A: They will only find scrambled ciphertext files (`.enc`). Without the unique 32-byte AES key (which is wrapped with an RSA key stored separately on the server), the stolen files cannot be opened.

8.2 Solutions to Common Problems

Problem	Cause	How to Fix It
"Invalid email or password"	Typo in email or password.	Check Caps Lock; re-type carefully; or ask an Admin to reset your password.
"Account has been deactivated"	Admin disabled the account.	Contact your institution's ExamGuard Administrator to reactivate your account.
Download button has a lock icon	The scheduled delivery date has not arrived yet.	Wait until the delivery date arrives (usually tomorrow or exam day). Authoring teachers can download their own papers immediately.
"Answer key cannot be uploaded before exam"	Today is earlier than the scheduled exam date.	Wait until the exam has taken place, then open the Answer Key modal.
"File size exceeds 10MB limit"	Attached PDF is too large.	Compress your PDF using standard PDF compression software before uploading.
Browser console shows `content.js:24` message	A browser extension (like Bitwarden or LastPass) had an idle background script.	This is completely harmless and does not affect ExamGuard. To silence it, open the app in an Incognito / InPrivate window.

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9. Security Tips for Everyday Use

1. **Keep Your Password Secret**: Never share your login credentials with colleagues, assistants, or students.
2. **Use Institutional Computers**: Always access ExamGuard from a secure university workstation. Avoid logging in from public internet cafes.
3. **Check the Green Verification Badge**: Before printing an exam paper, click **Verify Signature** to ensure the document seal is intact.
4. **Delete Local Downloads**: Once an examination paper is printed in the secure exam cell, delete the downloaded PDF from your computer's `Downloads` folder and empty your Recycle Bin.
5. **Always Sign Out**: When you finish working, click **Sign Out** in the top-right corner to clear your session token.

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10. Glossary of Simple Terms

- **AES-256**: A military-grade digital lock that scrambles a document into unreadable ciphertext.
- **Authentication**: Proving who you are by entering your email and password.
- **Ciphertext**: The scrambled, unreadable version of a file when it is encrypted.
- **Digital Signature**: A mathematical wax seal stamped on an encrypted file to prove who sent it and that it has not been altered.
- **Fail-Closed**: A safety design where the system immediately shuts down access if anything looks tampered with or corrupted.
- **JWT (Token)**: A temporary digital pass issued to your browser after you log in, valid for 24 hours.
- **Plaintext**: The original, readable PDF document before it is encrypted.
- **RBAC**: Restricting what pages and buttons you can see based on your job (`TEACHER`, `EXAM_COORDINATOR`, or `ADMIN`).

- **RSA-OAEP**: A secure asymmetric key wrapping system used to lock the AES key inside an armored envelope.
- **SHA-256**: A unique 64-character digital fingerprint of a document. If one comma changes, the fingerprint changes completely.
- **Verify-Before-Decrypt**: Checking that the fingerprint and digital seal are authentic *before* unlocking the safe.
- **Zeroization**: Overwriting sensitive key memory buffers with zeros as soon as decryption is finished.

End of ExamGuard Simple Step-by-Step User Manual.